

Normalisation

(every length $\div R$; wall plane at $\bar{w} = 0$, contact at $\bar{w} = 1$, so $\bar{w} \geq 1$)

$\bar{w}$	probe-centre position
$\bar{w}_s$	contact position, unknown; nominal 1 (surface itself at $\bar{w}_s - 1$)
$(\bar{w} - \bar{w}_s)$	gap between the two surfaces; 0 at contact
$\bar{a} = a/a_o$	motion gain, $\in (0, 1)$
b, p	shape parameters; $\theta = (b, p, \bar{w}_s)$ in both forms

Truth

(Brenner 1961 series, $\bar{a} = 1/c_\perp$)

$$\bar{a}(\bar{w}) = 1 - \frac{9}{8}\bar{w}^{-1} + \frac{1}{2}\bar{w}^{-3} - \frac{57}{100}\bar{w}^{-4} + \frac{1}{5}\bar{w}^{-5} + \frac{7}{200}\bar{w}^{-11} - \frac{1}{25}\bar{w}^{-12}$$

$$\text{near wall : } \bar{a} \simeq (\bar{w} - \bar{w}_s), \quad \text{far field : } 1 - \bar{a} \simeq \frac{9/8}{(\bar{w} - \bar{w}_s)}$$

Form A

(the proposal)

$$\bar{a}(\bar{w} - \bar{w}_s) = 1 - e^{-\frac{b}{p+1}(\bar{w} - \bar{w}_s)^{p+1}}$$

$$\bar{a}' = b(\bar{w} - \bar{w}_s)^p e^{-\frac{b}{p+1}(\bar{w} - \bar{w}_s)^{p+1}}$$

$$\begin{aligned} J_b &= \bar{a}' \left[\frac{1}{b} - \frac{(\bar{w} - \bar{w}_s)^{p+1}}{p+1} \right] \\ J_p &= \bar{a}' \left[\left(1 - \frac{b}{p+1}(\bar{w} - \bar{w}_s)^{p+1} \right) \ln(\bar{w} - \bar{w}_s) + \frac{b}{(p+1)^2}(\bar{w} - \bar{w}_s)^{p+1} \right] \\ J_{w_s} &= \bar{a}' \left[b(\bar{w} - \bar{w}_s)^p - \frac{p}{(\bar{w} - \bar{w}_s)} \right] \end{aligned}$$

Two limits of Form A

near wall, $\bar{a} \simeq \frac{b}{p+1}(\bar{w} - \bar{w}_s)^{p+1}$ must be linear in $(\bar{w} - \bar{w}_s)$:

$$p = 0, \quad b = 1$$

far field, $1 - \bar{a} = e^{-\frac{b}{p+1}(\bar{w} - \bar{w}_s)^{p+1}}$ must decay as a power; for $p+1 > 0$ it decays faster than every power:

$$p = -1$$

$$p = 0 \wedge p = -1 : \text{no solution, for any } (b, \bar{w}_s)$$

Form B

$$\bar{a}(\bar{w} - \bar{w}_s) = 1 - \left[\frac{b}{(\bar{w} - \bar{w}_s) + b} \right]^p$$

$$\bar{a}' = p b^p ((\bar{w} - \bar{w}_s) + b)^{-p-1}$$

$$\begin{aligned} J_b &= \bar{a}' \left[\frac{p}{b} - \frac{p+1}{(\bar{w} - \bar{w}_s) + b} \right] \\ J_p &= \bar{a}' \left[\frac{1}{p} + \ln \frac{b}{(\bar{w} - \bar{w}_s) + b} \right] \\ J_{w_s} &= \bar{a}' \frac{p+1}{(\bar{w} - \bar{w}_s) + b} \end{aligned}$$

Two limits of Form B

$$\bar{a} = 1 - \left[\frac{b}{(\bar{w} - \bar{w}_s) + b} \right]^p = 1 - \left(1 + \frac{(\bar{w} - \bar{w}_s)}{b} \right)^{-p}$$

near wall, $(\bar{w} - \bar{w}_s) \ll b$, drop the square and higher terms of $(1 + (\bar{w} - \bar{w}_s)/b)^{-p} \simeq 1 - p(\bar{w} - \bar{w}_s)/b$, then match $\bar{a} \simeq (\bar{w} - \bar{w}_s)$:

$$\bar{a} \simeq \frac{p}{b} (\bar{w} - \bar{w}_s) \quad \implies \quad \frac{p}{b} = 1$$

far field, $(\bar{w} - \bar{w}_s) \gg b$, drop the 1 against $(\bar{w} - \bar{w}_s)/b$, then match $1 - \bar{a} \simeq \frac{9/8}{(\bar{w} - \bar{w}_s)}$:

$$1 - \bar{a} \simeq \left(\frac{(\bar{w} - \bar{w}_s)}{b} \right)^{-p} = b^p (\bar{w} - \bar{w}_s)^{-p} \quad \implies \quad p = 1, \quad b = \frac{9}{8}$$

over-determined by 12.5% ($b = 1$ vs $b = \frac{9}{8}$), carried by one parameter

Under test

$$\theta[k+1] = \theta[k], \quad Q = 0$$

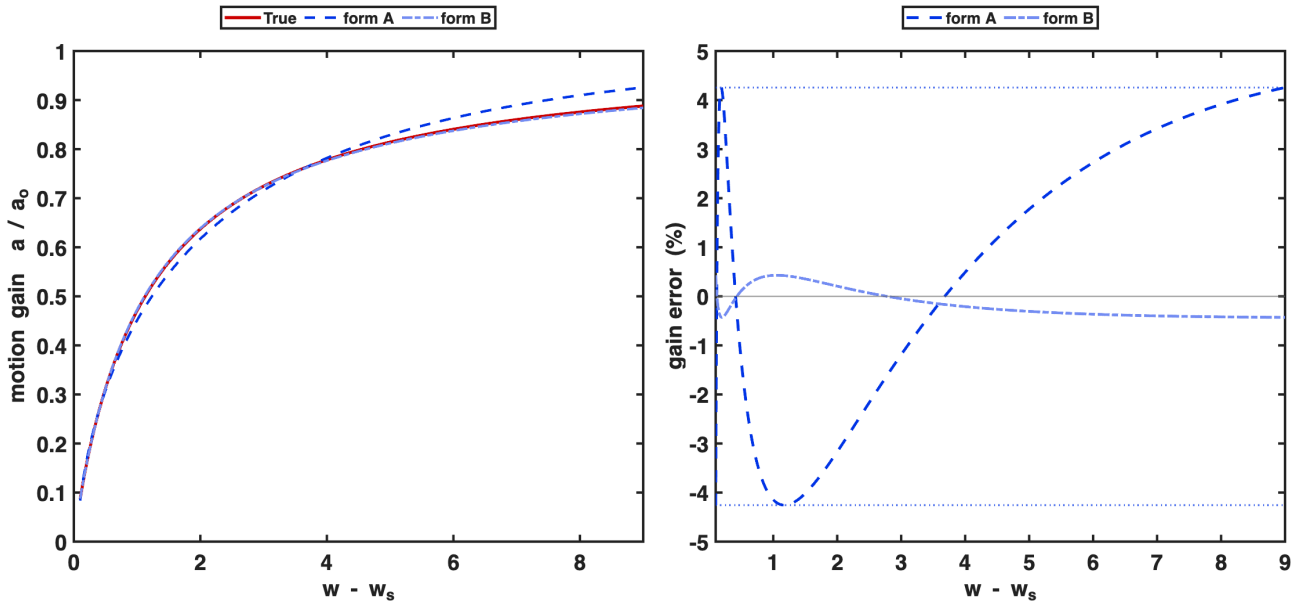
Best-case fit

($\bar{w} \in [1.1, 10]$, i.e. gap ($\bar{w} - \bar{w}_s$) $\in [0.1, 9]$; minimax, 2×10^4 samples)

$$\sup \left| \frac{\bar{a}_{\text{model}}}{\bar{a}_{\text{true}}} - 1 \right|$$

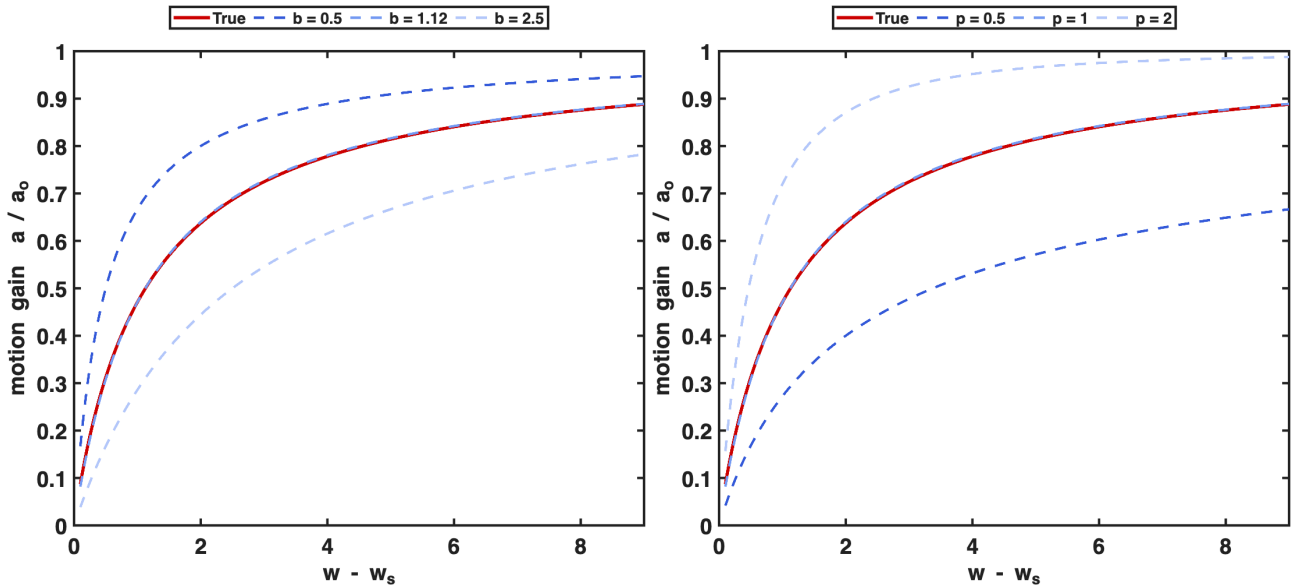
form	parameters	sup [%]	$\bar{w}_s - 1$ [μm]	equiosc. / starts
A	$b=0.406$, $p=-0.346$, $\bar{w}_s=1.0502$	4.256	+0.113	4/4, 6/6
B	$b=1.0566$, $p=0.9563$, $\bar{w}_s=0.9936$	0.429	-0.015	4/4, 5/5

Form B against its derived values: $b=1.0566$ vs $9/8 = 1.125$, $p=0.9563$ vs 1.



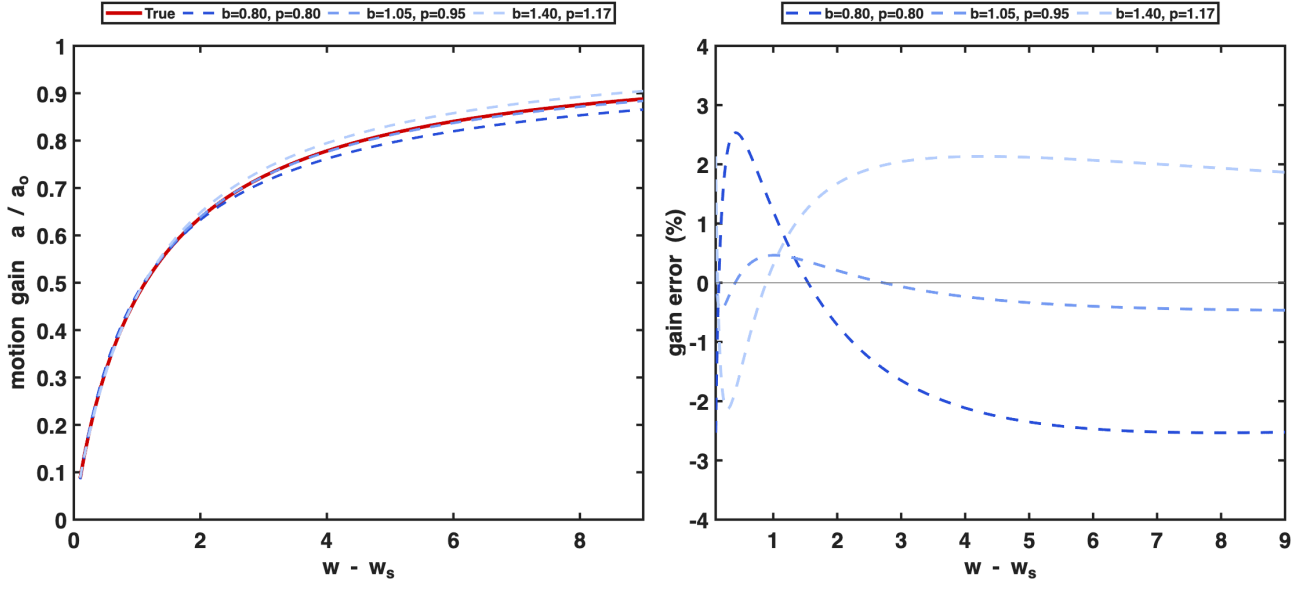
What b and p do

(Form B; left $p = 1$ with b swept, right $b = 9/8$ with p swept; red is the truth, the middle curve is the derived value)



The two are not separable

(three parameter sets from the floor of the error valley, each with its own best $\bar{w}_s$)



b	p	$\bar{w}_s$	p/b	sup [%]
0.80	0.8010	1.0060	1.0013	2.533
1.05	0.9528	0.9941	0.9074	0.464
1.40	1.1704	0.9837	0.8360	2.133

$b : 1.75\times,$ $p : 1.46\times,$ gain error : $< 2.6\%$, $\text{corr}(\ln b, \ln p) = 0.986$